Edge Intelligence Lab 4

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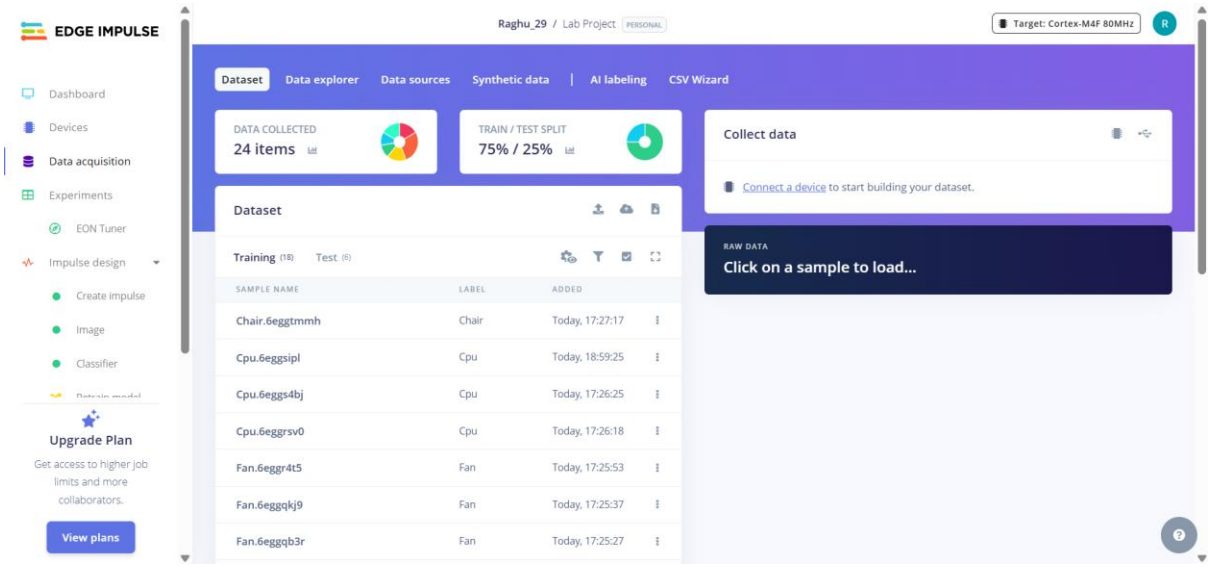
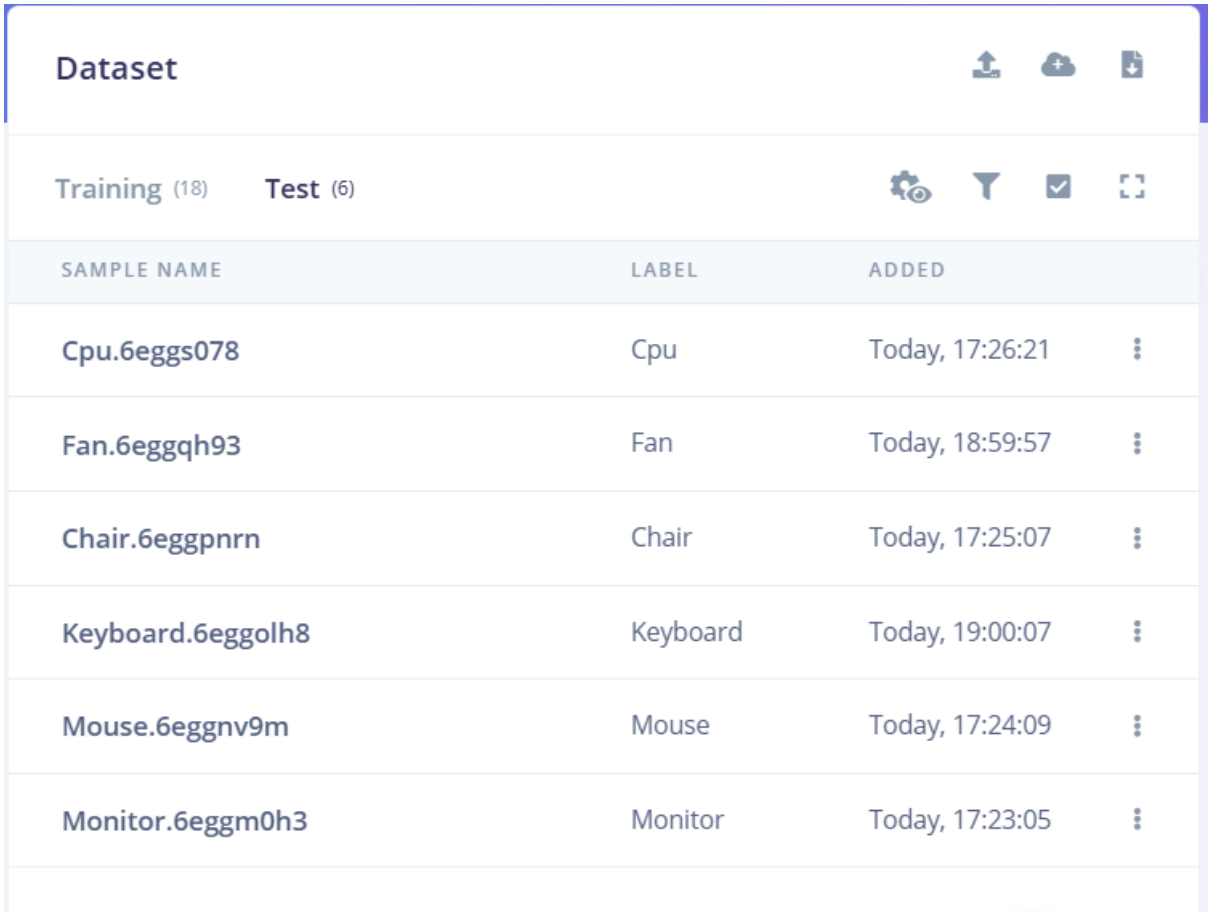
25MML0029

Edge Impulse

1.Dataset(Training)



Dataset(Testing)

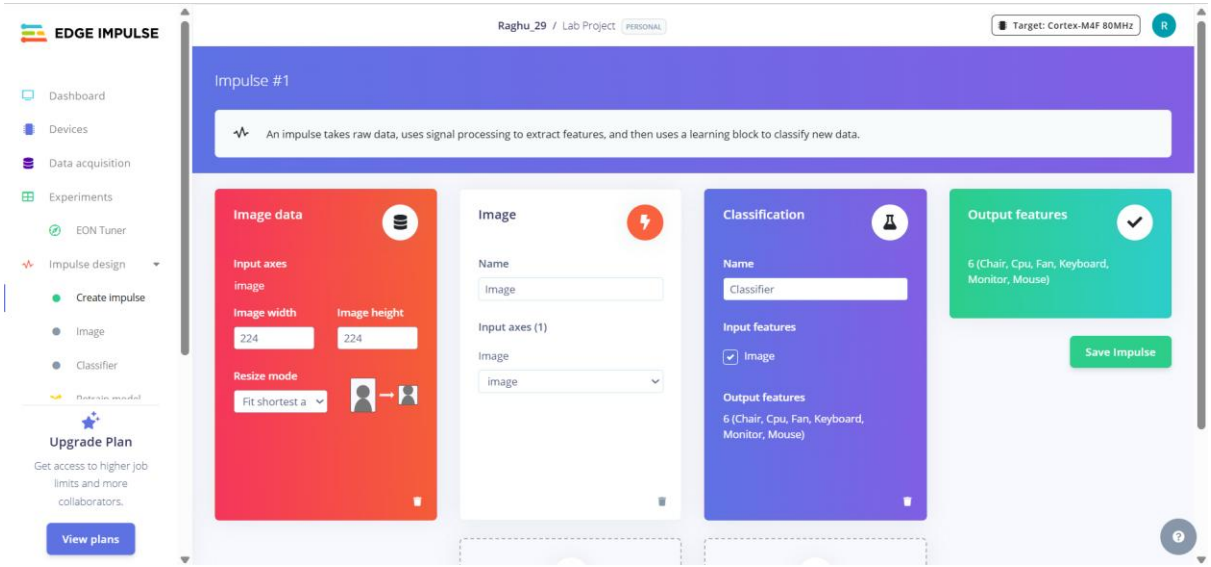


Dataset

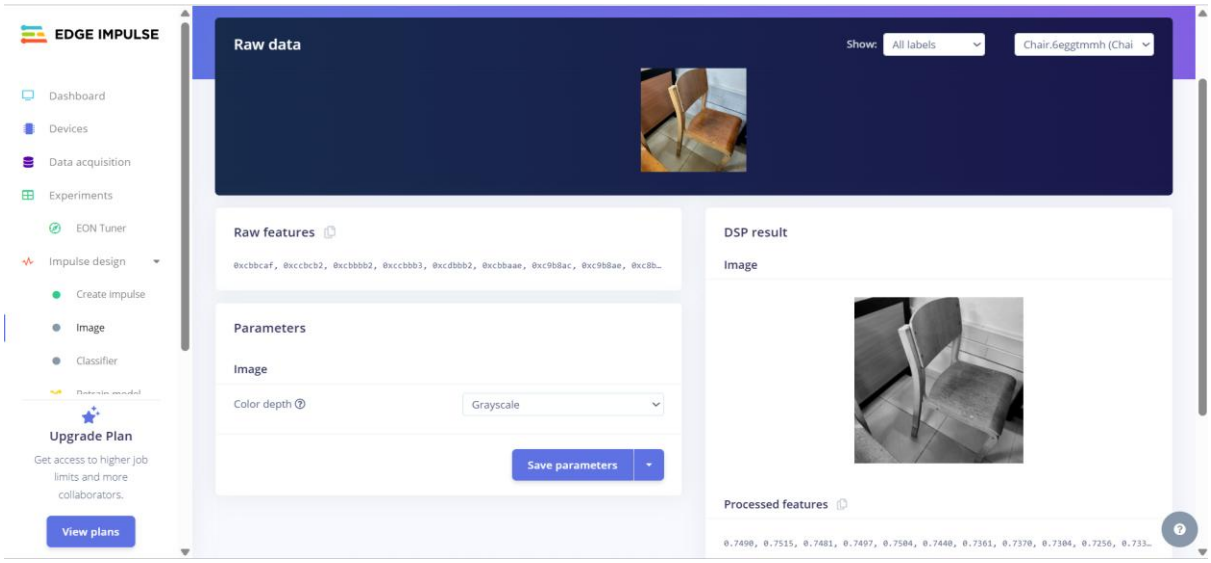
Training (18) Test (6)

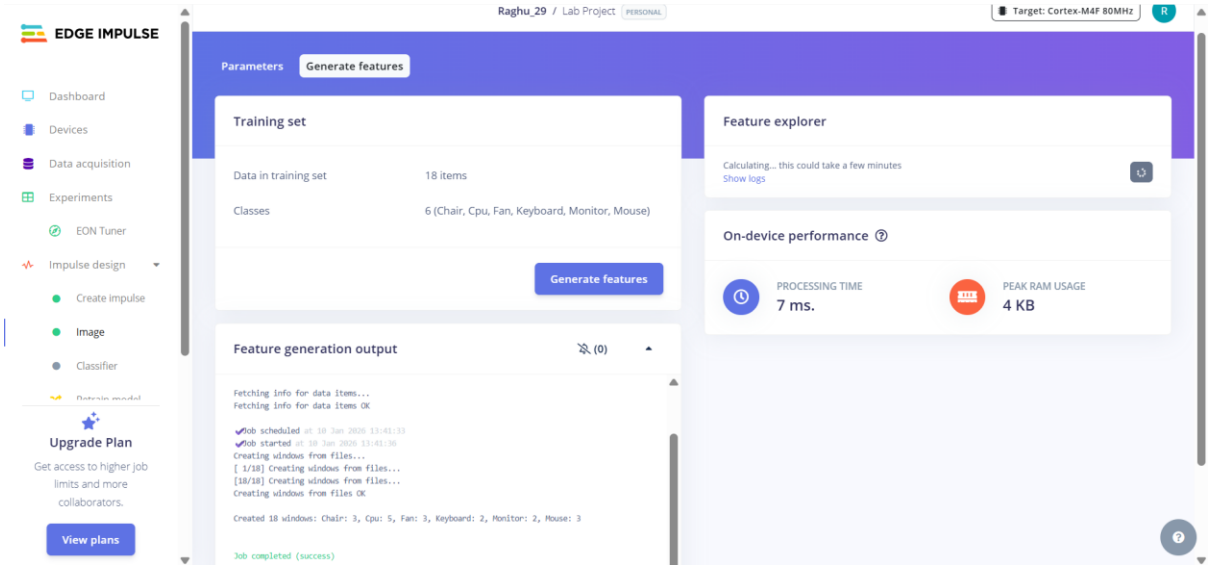
SAMPLE NAME	LABEL	ADDED
Cpu.6eggs078	Cpu	Today, 17:26:21
Fan.6eggqh93	Fan	Today, 18:59:57
Chair.6eggpnrn	Chair	Today, 17:25:07
Keyboard.6eggolh8	Keyboard	Today, 19:00:07
Mouse.6eggnv9m	Mouse	Today, 17:24:09
Monitor.6eggm0h3	Monitor	Today, 17:23:05

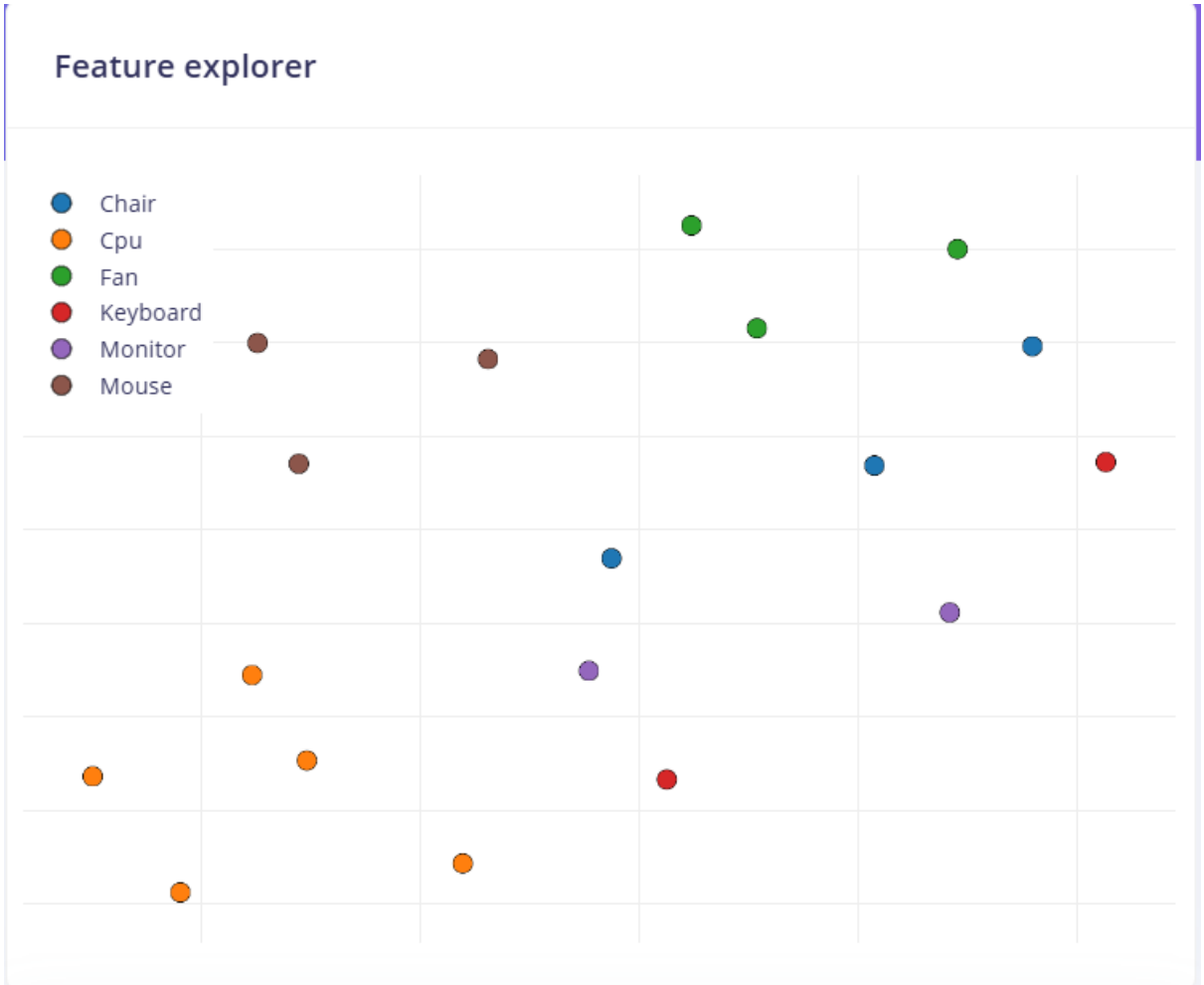
2.Creating Impulse



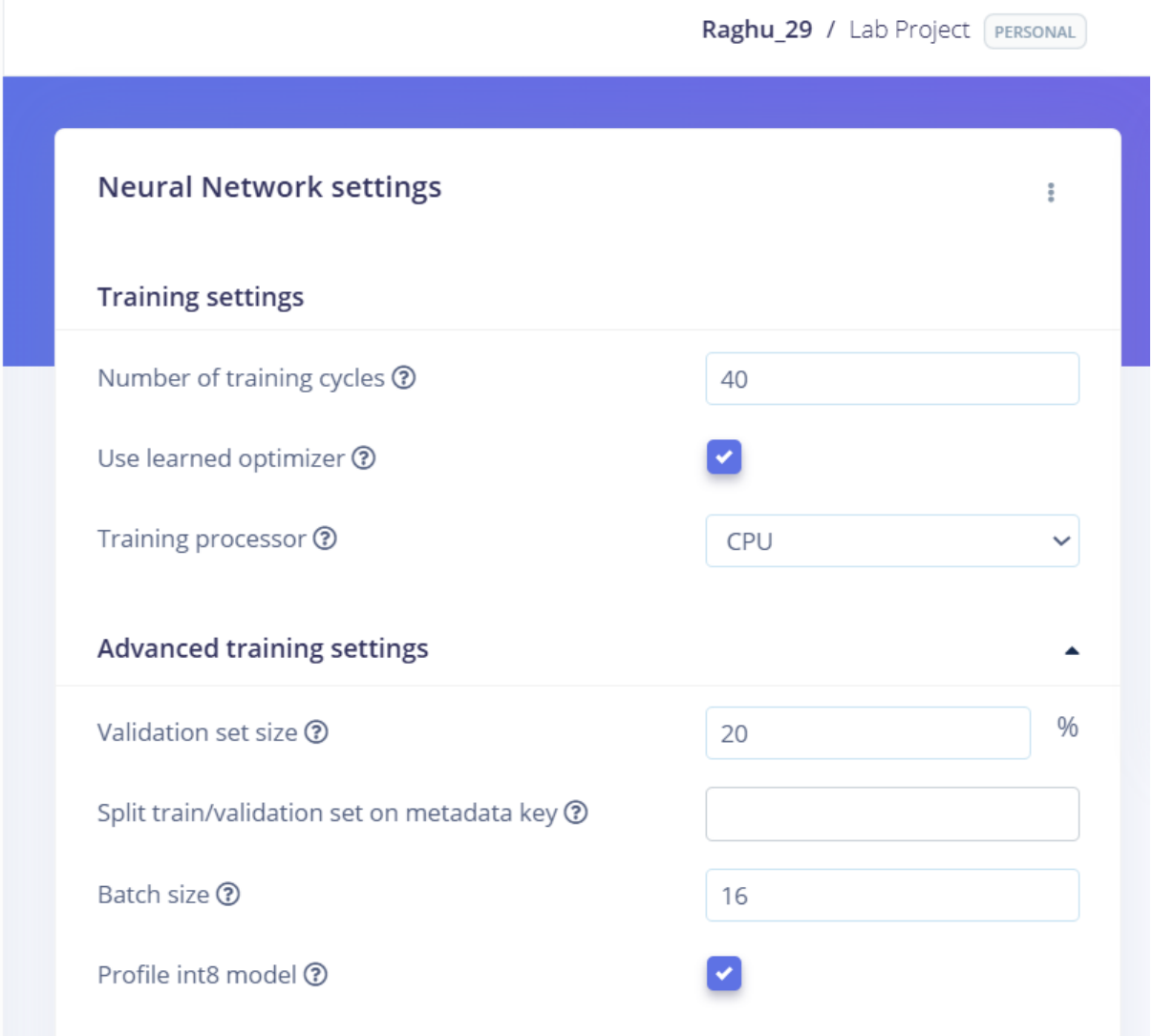
3.Image


EDGE IMPULSE
Dashboard
Devices
Data acquisition
Experiments
EON Tuner
Impulse design
Create impulse
Image
Classifier
Upgrade Plan
Get access to higher job limits and more collaborators.
View plans
Raw data
Show:
All labels
Chair.6eggtmmh (Chai
Raw features
0xcbbcaf, 0xccbcb2, 0xcbbbb2, 0xccbbb3, 0xcdbbb2, 0xcbbaae, 0xc9b8ac, 0xc9b8ae, 0xc8b…
Parameters
Image
Color depth
Grayscale
Save parameters
DSP result
Image
Processed features
0.7490, 0.7515, 0.7481, 0.7497, 0.7504, 0.7440, 0.7361, 0.7370, 0.7304, 0.7256, 0.733…


EDGE IMPULSE
Raghu_29 / Lab Project PERSONAL
Target: Cortex-M4F 80MHz
Dashboard
Devices
Data acquisition
Experiments
EON Tuner
Impulse design
Create impulse
Image
Classifier
Upgrade Plan
Get access to higher job limits and more collaborators.
View plans
Parameters
Generate features
Training set
Data in training set
18 items
Classes
6 (Chair, Cpu, Fan, Keyboard, Monitor, Mouse)
Generate features
Feature generation output
(0)
Fetching info for data items...
Fetching info for data items OK
Job scheduled at 10 Jan 2026 13:41:33
Job started at 10 Jan 2026 13:41:36
Creating windows from files...
[1/18] Creating windows from files...
[18/18] Creating windows from files...
Creating windows from files OK
Created 18 windows: Chair: 3, Cpu: 5, Fan: 3, Keyboard: 2, Monitor: 2, Mouse: 3
Job completed (success)
Feature explorer
Calculating... this could take a few minutes
Show logs
On-device performance
PROCESSING TIME
7 ms.
PEAK RAM USAGE
4 KB


Feature explorer
Chair
Cpu
Fan
Keyboard
Monitor
Mouse

4.Classifier



Neural network architecture

Neural network | Transfer learning

Input layer (50,176 features)

2D conv / pool layer (32 filters, 3 kernel size, 1 layer)

2D conv / pool layer (64 filters, 3 kernel size, 1 layer)

Flatten layer

Dense layer (128 neurons)

Dropout (rate 0.3)

Add an extra layer

Output layer (6 classes)

5.Training Images

```
Training on 14 inputs, validating on 4 inputs
Using batch size: 16
Loading velo checkpoint from checkpoint_path [/app/transfer-learning-weights/velo/aug12_continue_on_bigger
_2xbs_200kstep_bigproblem_v2_5620/params] with config_path [None]
Epoch 0: train loss 1.79217 validation loss 2.05238
Epoch 1: train loss 1.59873 validation loss 1.75943
Epoch 2: train loss 1.46760 validation loss 4.36739
Epoch 3: train loss 2.56794 validation loss 3.87023
Epoch 4: train loss 2.71738 validation loss 1.63957
Epoch 5: train loss 1.54458 validation loss 2.29701
Epoch 6: train loss 1.19537 validation loss 1.92388
Epoch 7: train loss 0.83239 validation loss 2.79517
Epoch 8: train loss 0.71776 validation loss 2.43153
Epoch 9: train loss 0.70691 validation loss 1.86232
Epoch 10: train loss 0.45562 validation loss 2.25635
Epoch 11: train loss 0.15258 validation loss 2.06584
Epoch 12: train loss 0.19826 validation loss 3.68694
Epoch 13: train loss 0.13391 validation loss 3.14898
Epoch 14: train loss 0.03829 validation loss 2.98332
Epoch 15: train loss 0.31633 validation loss 4.33078
```

```
Epoch 25: train loss 0.00017 validation loss 8.36901
Epoch 26: train loss 0.00001 validation loss 8.61195
Epoch 27: train loss 0.00001 validation loss 9.39276
Epoch 28: train loss 0.00000 validation loss 10.08535
Epoch 29: train loss 0.00000 validation loss 10.78537
Epoch 30: train loss 0.00000 validation loss 11.45444
Epoch 31: train loss 0.00000 validation loss 12.10588
Epoch 32: train loss 0.00000 validation loss 12.73097
Epoch 33: train loss 0.00000 validation loss 13.39288
Epoch 34: train loss 0.00000 validation loss 13.94639
Epoch 35: train loss 0.00000 validation loss 14.52792
Epoch 36: train loss 0.00000 validation loss 14.96626
Epoch 37: train loss 0.00000 validation loss 15.35581
Epoch 38: train loss 0.00000 validation loss 15.61591
Epoch 39: train loss 0.00000 validation loss 15.86794
Finished training

Saving best performing model... (based on validation loss)
Still saving model...
Still saving model...
```

```
Creating embeddings OK (took 1 second)

Calculating performance metrics...
Calculating inferencing time...
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
Calculating inferencing time OK
Calculating float32 accuracy...
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
Extracting TensorBoard logs...
Extracting TensorBoard logs OK

Model training complete

Model training complete

Model training complete

Model training complete

Job completed (success)
```

6.Model Accuracy

Model

Model version: Quantized (int8)

Last training performance (validation set)

ACCURACY
75.0%

LOSS
1.64

Confusion matrix (validation set)

	CHAIR	CPU	FAN	KEYBOARD	MONITOR	MOUSE
CHAIR	-	-	-	-	-	-
CPU	0%	100%	0%	0%	0%	0%
FAN	-	-	-	-	-	-
KEYBOARD	-	-	-	-	-	-
MONITOR	0%	100%	0%	0%	0%	0%
MOUSE	0%	0%	0%	0%	0%	100%
F1 SCORE		0.80			0.00	1.00

Metrics (validation set)

METRIC	VALUE
Weighted average Precision	0.58
Weighted average Recall	0.75
Weighted average F1 score	0.65